

Mini-Course on Generative AI Applications

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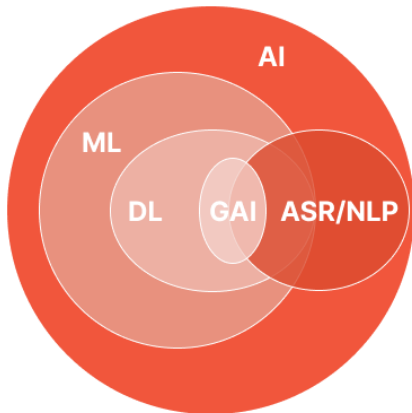
Session 1

- ▶ Introduction to LLMs and Transformers
- ▶ An Overview of Applications

Oct 28

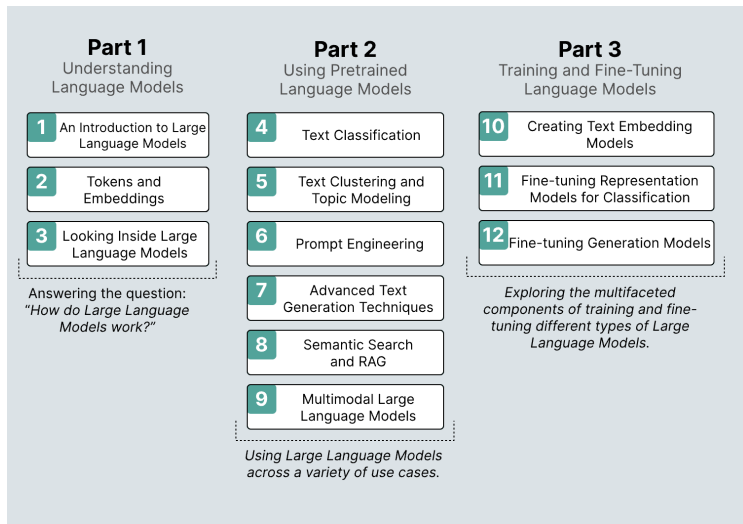
Introduction to LLMs and Transformers

Some Related Concepts



- **AI** - Artificial Intelligence
- **ML** - Machine Learning
- **DL** - Deep Learning
- **ASR/NLP** - Automatic Speech Recognition, Natural Language Processing
- **GAI** - Generative AI

LLMs Big Picture



Transformer Architectures

- ▶ Example 1: <https://bbycroft.net/llm>
- ▶ Example 2: <https://poloclub.github.io/transformer-explainer/>
- ▶ Example 3: <https://ig.ft.com/generative-ai/>

▶ **Multimodal vs. Single-Modal Models**

- ▶ Multimodal: work with multiple data types at the same time
- ▶ Examples: GPT-4o
- ▶ Single-Modal: GPT-3.5, BERT

▶ **Compact vs. Frontier Models**

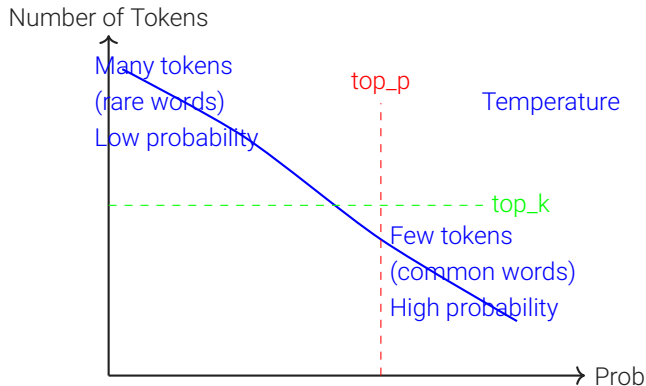
- ▶ **Compact:** Smaller, faster, cheaper
- ▶ Examples: Claude Haiko 3.5, GPT 4omini
- ▶ **Frontier:** Larger, more powerful, more expensive
- ▶ Examples: GPT-4, Claude Oppus

▶ **Paid vs. Open-Source Models**

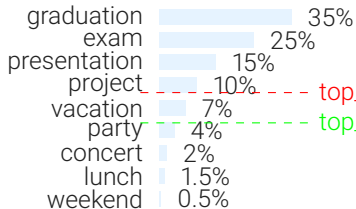
- ▶ Paid: ChatGPT, Claude
- ▶ Open-Source: LLaMA, Mistral

Temperature, top_p, and top_k

- ▶ **Temperature:** Controls randomness in output.
 - ▶ Lower (e.g., 0.1): More deterministic
 - ▶ default (usually half)
 - ▶ Higher (e.g., 0.9): More random
- ▶ **top_p (nucleus sampling):** Limits choices to top cumulative probabilities.
 - ▶ Example: top_p = 0.9 considers tokens until 90% cumulative probability is reached.
- ▶ **top_k:** Limits choices to top-k most likely next tokens.
 - ▶ Example: top_k = 50 considers only the 50 most probable next tokens.



Example: "The students were excited about their upcoming ____"



top_p (0.85): up to 85% cumulative probability

top_k (5): only top 5 tokens

Temperature effects:

- Low (0.3): High chance of "graduation" or "exam"
- Medium (0.6): More variety, might include "presentation" or "project"
- High (0.9): Could pick any option,

top_p (0.85):

Cumulative probability cutoff

top_k (5):

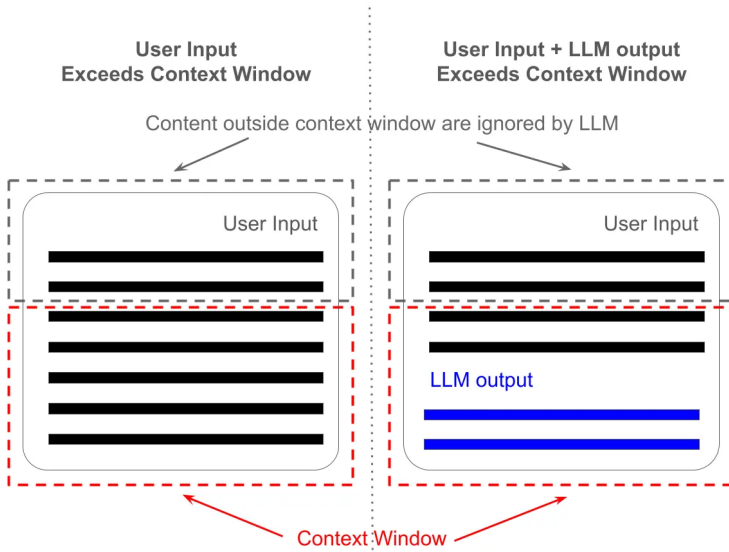
Only considers first 5 tokens

<https://platform.openai.com/playground/>

The screenshot displays the OpenAI Playground interface. On the left is a sidebar with navigation links: PLAYGROUND, Chat (selected), Realtime, Assistants, TTS, Completions, Cookbook, Forum, and Help. The main area is titled 'Chat' and contains a 'System instructions' field with a plus icon. Below this is a large text input area for the user's message, with a placeholder 'Enter user message...'. At the bottom of this area are buttons for 'User', 'Add', 'Run', and a keyboard shortcut icon. On the right side, there is a configuration panel with the following sections: 'Model' (set to gpt-4o), 'Functions' (with an '+ Add' button), 'Response format' (set to text), 'Model configuration' (with sliders for Temperature at 1.00 and Maximum length at 2048), 'Stop sequences' (with a text input field), 'Top P' (set to 1.00), 'Frequency penalty' (set to 0.00), and 'Presence penalty' (set to 0.00). At the bottom of the configuration panel is a 'Save as preset' button. The top of the interface shows the user 'Ali at IU' and a 'sample' dropdown, along with navigation links for Playground, Dashboard, Docs, and API reference.

Resource: <https://platform.openai.com/docs/examples>

Context Window



- ▶ **Retrieval-Augmented Generation (RAG):**

- ▶ Combines retrieval with generation
- ▶ $P(y|x) = \sum_{z \in Z} P(y|x, z)P(z|x)$

- ▶ **Non-RAG Models:**

- ▶ Generate based solely on training data
- ▶ $P(y|x) = f_{\theta}(x)$

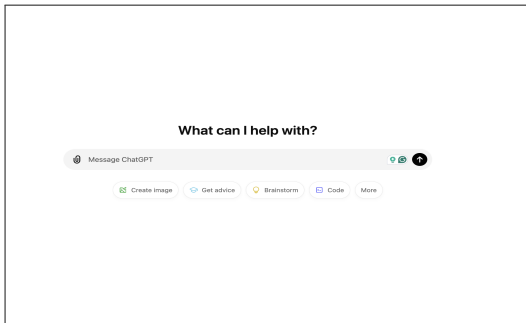
- ▶ **Fine-Tuning:**

- ▶ Modifies model parameters θ
- ▶ $\theta_{new} = \theta_{old} - \alpha \nabla L(\theta)$

APIs vs. Platforms

Platforms:

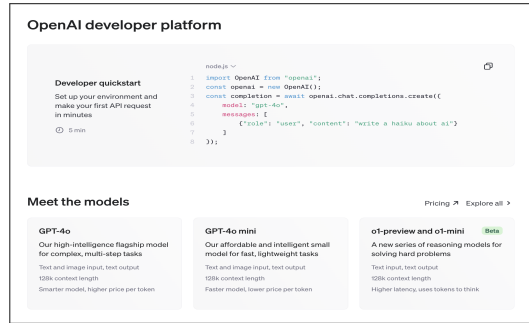
- ▶ User Friendly
- ▶ Good for many tasks



→ Third-Party Integration

APIs:

- ▶ Many features
- ▶ Higher Flexibility



Some Resources for LLMs

Educational Channels

▶ IBM Youtube

- ▶ youtube.com/@IBMTechology

▶ 3Blue1Brown

- ▶ 3blue1brown.com
- ▶ youtube.com/3blue1brown
- ▶ Excellent transformer series

▶ Hugging Face

- ▶ huggingface.co
- ▶ Courses, models & documentation
- ▶ transformers' pipelines

Repositories

- ▶ <https://github.com/jihoo-kim/awesome-production-llm>
- ▶ <https://github.com/Hannibal046/Awesome-LLM>
- ▶ <https://github.com/mlabonne/llm-course>
- ▶ <https://github.com/MalayAgr/generative-ai-with-llms-notes/tree/main>
- ▶ <https://llm-course.github.io/>

LLM Frameworks

- ▶ langchain.com
- ▶ llamaindex.ai

Monitoring, Logging& Observability

- ▶ langsmith.com
- ▶ langfuse.com
- ▶ helicone.ai
- ▶ logfire.dev

An Overview of Applications

Examples

- ▶ **GPT4 - GPT4o - GPT4omini - o1 preview - o1 mini** $\Rightarrow$ OpenAI
- ▶ **Calude 3.5 (Haiku, Sonnet, Opous)** $\Rightarrow$ Anthropic
- ▶ Gemini $\Rightarrow$ Google
- ▶ Llama $\Rightarrow$ Meta
- ▶ Mistral $\Rightarrow$ Mistral AI

Examples of search

- ▶ Perplexity
- ▶ **Phind**
- ▶ SearchGPT
- ▶ ...

Examples of research assistance

- ▶ **Elicit**
- ▶ Scite
- ▶ liner
- ▶ Consensus
- ▶ ...

Data Preprocessing and Cleaning

- ▶ Data Cleaning
- ▶ Data Visualization
- ▶ Data Handling
- ▶ Data Conversion

AI-powered Coding

- ▶ Cursor
- ▶ Copilot
- ▶ CodeGPT
- ▶ Many more

Tips & Tricks (1)

Data Handling

- ▶ Use of multi modality
- ▶ Large Datasets

Example Prompt 1

```
"Read the attached data and give me every possible basic and sophisticated analysis ....:
```

Example Prompt 2

```
"Suggest methods and best practices for cleaning and preprocessing this messy dataset. ....:
```

Some examples: Prompts for data analysis

Assisted Writing and Editing

- ▶ Grammar and style checking and changing
- ▶ $\text{\LaTeX}$ Assistance
 - ▶ TikZ Figures
- ▶ **Providing critical feedback**

Tips & Tricks (2)

Critical Feedback

- ▶ Role of LLM: Act as a domain expert
- ▶ Context Importance: Provide full paper details

Example Prompt

Role: Act as an Econometrician or (Reviewer for a top economics journal)

Context: I will provide a complete research paper for review

Task: Give me your review for ... in full LaTeX code.

"Read the paper and familiarize yourself with the content, models and findings, tone, style of writing ...:

1. Give me some feedback
2. Suggest improvements

Format the review with `\textcolor{red}{...}` for major concerns,
`\textcolor{blue}{...}` for minor revisions" (or say color code it)

Some Examples

- ▶ Sentiment Analysis
- ▶ Topic Modeling
 - ▶ Supervised and Unsupervised
- ▶ Classifications
- ▶ Regressions
- ▶ Relationship Extraction

Examples

- ▶ Chatbot Surveys
 - ▶ An Example: Yasna
- ▶ AI agents
- ▶ HAI
- ▶ Many more

Tips & Tricks (3)

Innovative Idea

- ▶ Claude Artifact
- ▶ GPT4o with Canvas

Example Prompt

can you design a simple demand and supply framework and that price is not in the equilibrium and step by step based on the forces of the market it gets to the equilibrium.

Example 1- Market Eq.

Example 2 - DA Market

Some Considerations

- ▶ Verifying accuracy in AI-generated outputs
- ▶ Disclosing AI assistance
- ▶ Data privacy (**For some tasks**)
- ▶ Knowing about bias in AI models
- ▶ Reproducibility

Thank you!